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THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee (the Committee) conferred upon it by international conventions for the prevention and control of marine pollution,

RECALLING ALSO that, at its sixty-second session, the Committee adopted, by resolution MEPC.203(62), amendments to the Annex of the Protocol of 1997 to amend the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that regulation 21 (Required EEDI) of MARPOL Annex VI, as amended, requires

of 1978 relating thereto (inclusion of regulations on energy efficiency for ships in MARPOL Annex VI),

NOTING that regulation 21 (Required EEDI) of MARPOL Annex VI, as amended, requires reference lines to be established for each ship type to which regulation 21 is applicable,

HAVING CONSIDERED, at its sixty-third session, the draft Guidelines for calculation of reference lines for use with the Energy Efficiency Design Index (EEDI),

ADOPTS the Guidelines for calculation of reference lines for use with the Energy Efficiency Design Index (EEDI), as set out at annex to the present resolution.

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GUIDELINES FOR CALCULATION OF REFERENCE LINES FOR USE WITH THE ENERGY EFFICIENCY DESIGN INDEX (EEDI)

1 The reference lines are established for each ship type to which regulation 21 (Required EEDI) of MARPOL Annex VI is applicable. The purpose of the EEDI is to provide

ENERGY EFFICIENCY DESIGN INDEX (EEDI)

1 The reference lines are established for each ship type to which regulation 21 (Required EEDI) of MARPOL Annex VI is applicable. The purpose of the EEDI is to provide a fair basis for comparison, to stimulate the development of more efficient ships in general and to establish the minimum efficiency of new ships depending on ship type and size. Hence, the reference lines for each ship type is calculated in a transparent and robust manner.

2 Ship types are defined in regulation 2 of MARPOL Annex VI. The reference line for each ship type is used for the determination of the required EEDI as defined in regulation 21 of MARPOL Annex VI.

3 These guidelines apply to the following ships types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier. It is noted that a method of calculating reference lines has not been established for ro-ro cargo ships, ro-ro cargo ships (vehicle), ro- ro-passenger ships and passenger ships, as well as for ships with diesel-electric propulsion, turbine propulsion and hybrid propulsion.

Definition of a reference line

ships, ro-ro cargo ships (vehicle), ro- ro-passenger ships and passenger ships, as well as for ships with diesel-electric propulsion, turbine propulsion and hybrid propulsion.

Definition of a reference line

4 A reference line is defined as a curve representing an average index value fitted on a set of individual index values for a defined group of ships.

5 One reference line is developed for each ship type to which regulation 21 of MARPOL Annex VI is applicable, ensuring that only data from comparable ships are included in the calculation of each reference line.

6 The reference line value is formulated as Reference line value = a (100% deadweight) -c where "a" and "c" are parameters determined from the regression curve fit.

7 Input data for the calculation of the reference lines is filtered through a process where data deviating more than two standard deviations from the regression line are discarded. The regression is then applied again to generate a corrected reference line. For the purpose of documentation, discarded data is listed with the ships IMO number.

Data sources

discarded. The regression is then applied again to generate a corrected reference line. For the purpose of documentation, discarded data is listed with the ships IMO number.

Data sources

8 IHS Fairplay (IHSF) database is selected as the standard database delivering the primary input data for the reference line calculation. For the purpose of the EEDI reference line calculations, a defined version of the database is archived as agreed between the Secretariat and IHSF.

9 For the purpose of calculating the reference lines, data relating to existing ships of 400 GT and above from the IHS F database delivered in the period from 1 January 1999 to 1 January 2009 are used.

10 T he following data from the IHSF database on ships with conventional propulsion systems is used when calculating the reference lines:

.1 data on the ships' capacity is used as Capacity for each ship type as defined in MEPC.XXX(63);

10 T he following data from the IHSF database on ships with conventional propulsion systems is used when calculating the reference lines:

.1 data on the ships' capacity is used as Capacity for each ship type as defined in MEPC.XXX(63);

.2 data on the ships' service speed is used as reference speed Vref; and

.3 data on the ships' total installed main power is used as MCRME(i).

11 For some ships, some data entries may be blank or contain a zero (0) in the database. Datasets with blank power, capacity and/or speed data should be removed from the reference line calculations. For the purpose of later references, the omitted ships should be listed with their IMO number.

12 To ensure a uniform interpretation, the association of ship types defined in regulation 2 of MARPOL Annex VI, with the ship types given by the IHSF database and defined by the so-called Stat codes, is shown in the appendix to this guideline. Table 1 in the appendix lists the ship types from IHSF used for the calculation of reference lines. Table 2 lists the IHSF ship types not used when calculating the reference lines.

Calculation of reference lines

appendix lists the ship types from IHSF used for the calculation of reference lines. Table 2 lists the IHSF ship types not used when calculating the reference lines.

Calculation of reference lines

13 To calculate the reference line, an estimated index value for each ship contained in the set of ships per ship type is calculated using the following assumptions:

.1 the carbon emission factor is constant for all engines, i.e. CF,ME = CF,AE = CF = 3.1144 g CO2/g fuel;

.2 t he specific fuel consumption for all ship types is constant for all main engines, i.e. SFCME = 190 g/kWh;

.3 PME(i) is 75% of the total installed main power (MCRME(i));

.4 the specific fuel consumption for all ship types is constant for all auxiliary engines, i.e. SFCAE = 215 g/kWh;

.5 PAE is the auxiliary power and is calculated according to paragraphs 2.5.6.1 and 2.5.6.2 of the annex to MEPC.XXX(63);

.6 no correction factors are used; and

.7 innovative mechanical energy efficiency technology, shaft motors and other innovative energy efficient technologies are all excluded from the reference line calculation, i.e. PAEeff = 0, PPTI = 0, Peff = 0.

.7 innovative mechanical energy efficiency technology, shaft motors and other innovative energy efficient technologies are all excluded from the reference line calculation, i.e. PAEeff = 0, PPTI = 0, Peff = 0.

14 The equation for calculating the estimated index value for each ship (excluding containerships see paragraph 15) is as follows:

ref
NME
i
AEiME
VCapacity
PP
ValueIndexEstimated

1
215190
1144.3

15 For containerships, 70 per cent of the deadweight (70% DWT) is used as capacity for calculating the estimated index value for each containership as follows:

ref
NME
i
AEiME
V
PP
ValueIndexEstimated

DWT%70
215190
1144.3 1

Calculation of reference line parameters "a" and "c"

16 For all ship types to which these guidelines apply, parameters "a" and "c" are determined from a regression analysis undertaken by plotting the calculated estimated index values against 100 per cent deadweight (100% DWT).

Documentation

16 For all ship types to which these guidelines apply, parameters "a" and "c" are determined from a regression analysis undertaken by plotting the calculated estimated index values against 100 per cent deadweight (100% DWT).

Documentation

17 For purposes of transparency, the ships used in the calculation of the reference lines should be listed with their IMO numbers and the numerator and denominator of the index formula, as given in paragraphs 14 and 15. The documentation of the aggregated figures preserves the individual data from direct access but offers sufficient information for possible later scrutiny.

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APPENDIX

1 To ensure a uniform interpretation, ship types defined in regulation 2 of MARPOL Annex VI are compared to the ship types given in the IHSF database.

2 The IHSF Stat code system provides several levels of definition as follows:

.1 Highest level:

A Cargo carrying
B Work vessel
W Non-seagoing merchant ships

MARPOL Annex VI are compared to the ship types given in the IHSF database.

2 The IHSF Stat code system provides several levels of definition as follows:

.1 Highest level:

A Cargo carrying
B Work vessel
W Non-seagoing merchant ships
X Non-merchant
Y Non-propelled
Z Non-ship structures

For the purpose of the EEDI, only group " A cargo carrying" needs to be considered.
A graphical representation of this is given below.

.2 The next level comprises:

A1 Tankers
A2 Bulk carriers
A3 Dry cargo/passenger

There are further differentiations until level five, e.g. "A31A2GX General Cargo Ship", and each category is described.

The complete list is attached.

3 The ship types from the IHSF Stat code 5 (Statcode5v1075) used for the calculation of reference lines for the following ship types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier, are set out in Table 1.

reference lines for the following ship types: bulk carrier, gas carrier, tanker, containership, general cargo ship, refrigerated cargo carrier and combination carrier, are set out in Table 1. The IHSF database ship types, not used in the calculation of reference lines for the specific ship types, are set out in Table 2, e.g. ships built for sailing on the Great Lakes and landing craft.

Table 1: Ship types from IHSF used for the calculation of reference lines for use with the EEDI

.1 Bulk carrier		
Bulk dry	A21A2BC	Bulk carrier A single deck cargo vessel with an arrangement of topside ballast tanks for the carriage of bulk dry cargo of a homogeneous nature.
Bulk dry	A21B2BO	Ore carrier A single deck cargo ship fitted with two longitudinal bulkheads. Ore is carried in the centreline holds only.
Self-discharging bulk dry	A23A2BD	Bulk cargo carrier, self-discharging A bulk carrier fitted with self-trimming holds, a conveyor belt (or similar system) and a boom which can discharge cargo alongside or to shore without the assistance of any external equipment.
Other dry bulk	A24A2BT	Cement carrier A single deck cargo vessel fitted with pumping

conveyor belt (or similar system) and a boom which can discharge cargo alongside or to shore without the assistance of any external equipment.

Other dry bulk

A24A2BT Cement carrier A single deck cargo vessel fitted with pumping arrangements for the carriage of cement in bulk. There are no weather deck hatches. May be self-discharging.

A24B2BW Wood chips carrier, self-unloading A single deck cargo vessel with high freeboard for the carriage of wood chips. May be self-discharging.

A24C2BU Urea carrier A single deck cargo vessel for the carriage of urea in bulk. May be self-discharging.

A24D2BA Aggregates carrier A single deck cargo vessel for the carriage of aggregates in bulk. Also known as a sand carrier. May be self-discharging.

A24E2BL Limestone carrier A single deck cargo vessel for the carriage of limestone in bulk. There are no weather deck hatches. May be self-discharging.

.2 Gas carrier Liquefied gas

A11A2TN LNG tanker A tanker for the bulk carriage of liquefied natural gas (primarily methane) in independent insulated tanks. Liquefaction is achieved at temperatures down to -163 deg C.

A11B2TG LPG tanker

.2 Gas carrier Liquefied gas
A11A2TN LNG tanker
A tanker for the bulk carriage of liquefied natural gas (primarily methane) in independent insulated tanks. Liquefaction is achieved at temperatures down to -163 deg C.
A11B2TG LPG tanker
A tanker for the bulk carriage of liquefied petroleum gas in insulated tanks, which may be independent or integral. The cargo is pressurized (smaller vessels), refrigerated (larger vessels) or both ("semi-pressurized") to achieve liquefaction.
A11C2LC CO2 tanker A tanker for the bulk carriage of liquefied carbon dioxide.
A11A2TQ CNG tanker
A tanker for the bulk carriage of compressed natural gas. Cargo remains in gaseous state but is highly compressed.
.3 Tanker Chemical
A12A2LP Molten sulphur tanker A tanker for the bulk carriage of molten sulphur in insulated tanks at a high temperature.
A12A2TC Chemical tanker
A tanker for the bulk carriage of chemical cargoes, lube oils, vegetable/animal oils and other chemicals as defined in the International Bulk Chemical Code. Tanks are coated with suitable materials which are inert to the cargo.
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cargoes, lube oils, vegetable/animal oils and other chemicals as defined in the International Bulk Chemical Code. Tanks are coated with suitable materials which are inert to the cargo.
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A12B2TR Chemical/ products tanker A chemical tanker additionally capable of the carriage of clean petroleum products.
A12C2LW Wine tanker A cargo ship designed for the bulk transport of wine in tanks. Tanks will be stainless steel or lined. New vessels will be classified as chemical carriers.
A12D2LV Vegetable oil tanker A cargo ship designed for the bulk transport of vegetable oils in tanks. Tanks will be stainless steel or lined. New vessels will be classified as chemical carriers.
A12E2LE Edible oil tanker A cargo ship designed for the bulk transport of edible oils in tanks. Tanks will be stainless steel or lined. New vessels will be classified as chemical carriers.
A12F2LB Beer tanker A tanker for the bulk carriage of beer.

A cargo ship designed for the bulk transport of edible oils in tanks. Tanks will be stainless steel or lined. New vessels will be classified as chemical carriers.

A12F2LB Beer tanker A tanker for the bulk carriage of beer.

A12G2LT Latex tanker A tanker for the bulk carriage of latex.

A12H2LJ Fruit juice tanker A tanker for the bulk carriage of fruit juice concentrate in insulated tanks.

Oil

A13A2TV Crude oil tanker A tanker for the bulk carriage of crude oil.

A13A2TW Crude/oil products tanker A tanker for the bulk carriage of crude oil but also for carriage of refined oil products.

A13B2TP Products tanker A tanker for the bulk carriage of refined petroleum products, either clean or dirty.

A13B2TU Tanker (unspecified) A tanker whose cargo is unspecified.

A13C2LA Asphalt/ Bitumen tanker A tanker for the bulk carriage of asphalt/bitumen at temperatures between 150 and 200 deg C.

A13E2LD Coal/oil mixture tanker A tanker for the bulk carriage of a cargo of coal and oil mixed as a liquid and maintained at high temperatures.

Other liquids

A14A2LO Water tanker A tanker for the bulk carriage of water.

A13E2LD Coal/oil mixture tanker A tanker for the bulk carriage of a cargo of coal and oil mixed as a liquid and maintained at high temperatures.
Other
liquids
A14A2LO Water tanker A tanker for the bulk carriage of water.
A14F2LM Molasses tanker A tanker for the bulk carriage of molasses.
A14G2LG Glue tanker A tanker for the bulk carriage of glue.
A14H2LH Alcohol tanker A tanker for the bulk carriage of alcohol.
A14N2LL Caprolactam tanker A tanker for the bulk carriage of caprolactam, a chemical used in the plastics industry for the production of polyamides.
Chemical A12A2TL Parcels tanker
A chemical tanker with many segregated cargo
tanks to carry multiple grades of chemicals as
defined in the International Bulk Chemical Code.
Typically these can have between 10
and 60 different tanks.
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and 60 different tanks.
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.4 Containership Container A33A2CC Containership (fully cellular) A single deck cargo vessel with boxed holds fitted with fixed cellular guides for the carriage of containers.
.5 General
cargo ship
General
cargo A31A2GX General cargo ship
A single or multi-deck cargo vessel for the
carriage of various types of dry cargo. Single
deck vessels will typically have box-shaped
holds. Cargo is loaded and unloaded through
weather deck hatches.
Other dry
cargo A38H2GU Pulp carrier A vessel designed for carrying paper pulp.
.6 Refrigerated
cargo carrier
Refrigerated
cargo A34A2GR Refrigerated cargo ship A multi-deck cargo ship for the carriage of refrigerated cargo at various temperatures.
.7
Combination
carrier
Bulk dry/oil A22A2BB Bulk/oil carrier (OBO) A bulk carrier arranged for the alternative (but not simultaneous) carriage of crude oil.

.7
Combination
carrier
Bulk dry/oil A22A2BB Bulk/oil carrier (OBO) A bulk carrier arranged for the alternative (but not simultaneous) carriage of crude oil.
Bulk dry/oil A22B2BR Ore/oil carrier An ore carrier arranged for the alternative (but not simultaneous) carriage of crude oil.
Bulk dry/oil A22A2BP
Ore/bulk/
products
carrier
A bulk carrier arranged for the alternative (but
not simultaneous) carriage of oil products.

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Table 2: Ship types from IHSF not included in the calculation of
reference lines for use with the EEDI

.1 Bulk
carrier
Bulk dry A21A2BG Bulk carrier, laker only
A single deck cargo vessel
with dimensions suited to the
limitations of Great Lakes of
North America trade,
unsuitable for open sea
navigation. Hatches are
more numerous than
standard bulk carriers, and
much wider than they are
long.

		A single deck cargo vessel with dimensions suited to the limitations of Great Lakes of North America trade, unsuitable for open sea navigation. Hatches are more numerous than standard bulk carriers, and much wider than they are long.
Bulk dry	A21A2BV	Bulk carrier (with vehicle decks) A bulk carrier with movable decks for the additional carriage of new vehicles.
Bulk dry/oil	A22A2BB	Bulk/oil carrier (OBO) A bulk carrier arranged for the alternative (but not simultaneous) carriage of crude oil.
Bulk dry/oil	A22B2BR	Ore/oil carrier An ore carrier arranged for the alternative (but not simultaneous) carriage of crude oil.
Bulk dry/oil	A22A2BP	Ore/bulk/products carrier A bulk carrier arranged for the alternative (but not simultaneous) carriage of oil products.
Self-discharging bulk dry	A23A2BK	Bulk cargo carrier, self-discharging, laker A Great Lakes bulk carrier fitted with a conveyor belt (or similar system) and a boom which can discharge cargo alongside or to shore without the assistance of any external equipment.
Other bulk dry	A24H2BZ	Powder carrier A single deck cargo vessel for the carriage of fine powders such as fly ash.

boom which can discharge
cargo alongside or to shore
without the assistance of any
external equipment.
Other bulk dry A24H2BZ Powder carrier
A single deck cargo vessel
for the carriage of fine
powders such as fly ash.
There are no weather deck
hatches.
Other bulk dry A24G2BS Refined sugar carrier
A single deck cargo vessel
for the carriage of refined
sugar. Sugar is loaded in
bulk and bagged in transit
(BIBO Bulk In Bag Out).
.2 Gas
carrier Liquefied gas A11B2TH LPG/chemical tanker
An LPG tanker additionally
capable of the carriage of
chemical products as defined
in the International Bulk
Chemical Code.
.3 Tanker Oil A13A2TS Shuttle tanker
A tanker for the bulk carriage
of crude oil specifically for
operation between offshore
terminals and refineries. Is
typically fitted with bow
loading facilities.
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.4 Containership Container A33B2CP Passenger/ containership
A containership with
accommodation for the
carriage of more

.4 Containership Container A33B2CP Passenger/ containership
A containership with
accommodation for the
carriage of more
than 12 passengers.
.5 General
cargo ship
General cargo
A31A2GO Open hatch cargo ship
A large single deck cargo
vessel with full width hatches
and boxed holds for the
carriage of unitized dry cargo
such as forest products and
containers. Many are fitted
with a gantry crane.
General cargo A31A2GS
General
cargo/tanker
(container/oil/bulk
COB ship)
A general cargo ship with
reversible hatch covers; one
side is flush and the other is
fitted with baffles for use with
liquid cargoes. Containers
can be carried on the hatch
covers in dry cargo mode.
General cargo A31A2GT General cargo/tanker A general cargo ship fitted with tanks for the additional carriage of liquid cargo.
General cargo A31C2GD Deck cargo ship
A vessel arranged for
carrying unitized cargo on
deck only. Access may be
by use of a ro-ro ramp.
Passenger/general
cargo A32A2GF General cargo/ passenger ship
A general cargo ship with
accommodation for the

A vessel arranged for carrying unitized cargo on deck only. Access may be by use of a ro-ro ramp.

Passenger/general cargo A32A2GF General cargo/ passenger ship A general cargo ship with accommodation for the carriage of more than 12 passengers.

Other dry cargo A38A2GL Livestock carrier A cargo vessel arranged for the carriage of livestock.

Other dry cargo A38B2GB Barge carrier A cargo vessel arranged for the carriage of purpose built barges (lighters) loaded with cargo. Typically loading is by way of a gantry crane. Also known as Lighter Aboard SHip vessels (LASH).

Other dry cargo A38C3GH Heavy load carrier, semi-submersible A heavy load carrier which is semi-submersible for the float on loading/unloading of the cargoes.

Other dry cargo A38C3GY Yacht carrier, semi-submersible A semi-submersible heavy load carrier specifically arranged for the carriage of yachts.

Other dry cargo A38D2GN Nuclear fuel carrier A cargo vessel arranged to carry nuclear fuel in flasks.

Other dry cargo A38D2GZ Nuclear fuel carrier (with ro-ro facility) A nuclear fuel carrier which is loaded and unloaded by way of a ro-ro ramp.

Other dry cargo A38D2GN Nuclear fuel carrier A cargo vessel arranged to carry nuclear fuel in flasks.
Other dry cargo A38D2GZ Nuclear fuel carrier (with ro-ro facility) A nuclear fuel carrier which is loaded and unloaded by way of a ro-ro ramp.
Other dry cargo A38B3GB Barge carrier, semi-submersible
A barge carrier which is
semi-submersible for the
float on loading/unloading of
the barges.
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Other dry cargo A38C2GH Heavy load carrier
A cargo vessel able to carry
heavy and/or outsized
individual cargoes. Cargo
may be carried on deck or in
holds and may be loaded by
crane and/or ro-ro ramps.

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